

Oxidations and Reductions of Alcohols and Carbonyls

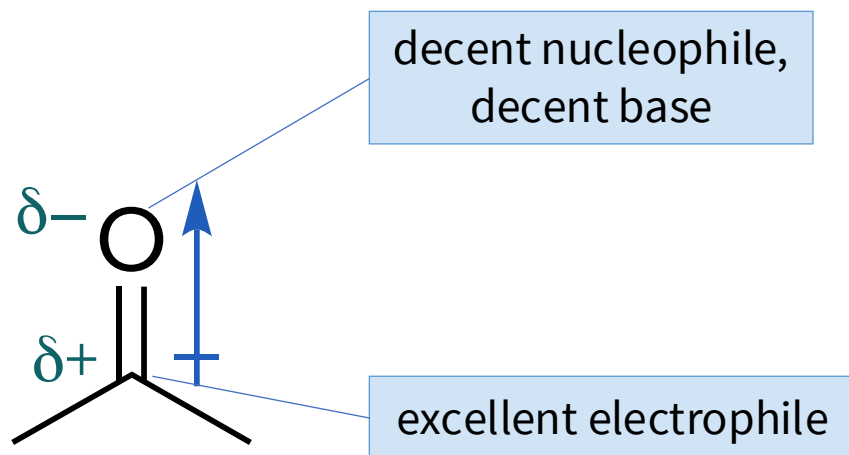
Organic Chemistry II

CHEM 2312

Dr. Michael Evans

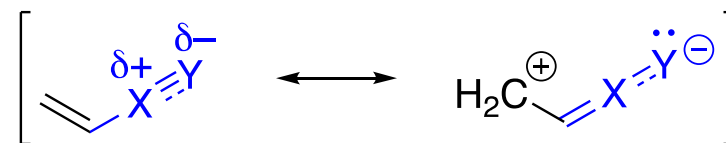
1. Describe the carbonyl group using resonance forms and dipole moment; recognize the carbonyl carbon as electrophilic.
2. Name carbonyl-containing functional groups; classify a given carbonyl compound as a ketohyde or carboxylic acid derivative (CAD).
3. Recognize oxidation and reduction of organic compounds as involving a change in oxidation number at carbon; apply oxidation ladders for carbonyl compounds and hydrocarbon groups.
4. Describe the mechanism of complex metal hydride reductions of ketohydes using elementary steps.
5. Predict the products of reduction reactions of ketohydes using complex metal hydride reagents.
6. Predict the products of strong and weak oxidations of alcohols; apply oxidation reactions of alcohols in synthesis.
7. Apply “nature’s redox agents,” including NADH, FADH₂, and QH₂ (and their oxidized forms) in biochemical redox reactions.

- The **carbonyl group** is a carbon–oxygen double bond
- Polarized toward oxygen, with δ^- on O and δ^+ on C
- Trigonal planar geometry at the carbonyl carbon



- The second best resonance form reveals reactivity

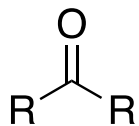
- Classic example of an electron-withdrawing group (EWG); recall the general structure of an EWG



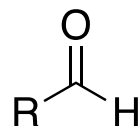
- Heteroatom-substituted carbonyl groups form three-atom conjugated systems



- Carbonyl-containing functional groups are named based on the types of atoms linked to the carbonyl carbon
- Ketones and aldehydes (“ketohydes”)

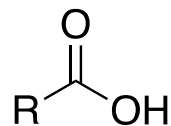


ketone

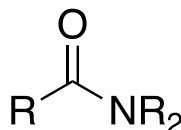


aldehyde

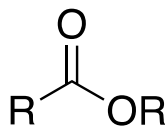
- Carboxylic acids and carboxylic acid derivatives (CADs)



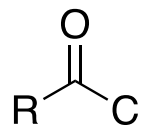
carboxylic acid



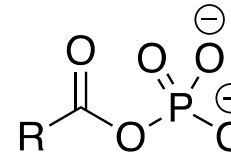
amide



ester

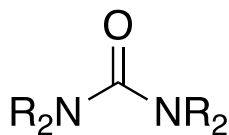


acyl chloride

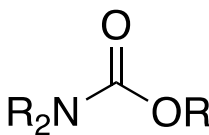


acyl phosphate

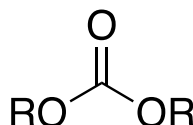
- Carbonic acid derivatives



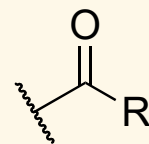
urea



carbamate

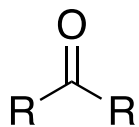


carbonate

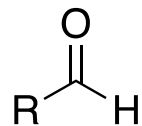


acyl group

- **Class I: Ketohydes.** These compounds do not have a leaving group linked to the carbonyl carbon

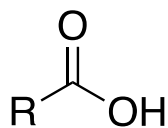


ketone

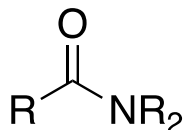


aldehyde

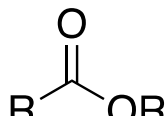
- **Class II: Carboxylic Acid Derivatives.** These compounds do have a leaving group linked to the carbonyl carbon



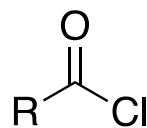
carboxylic acid



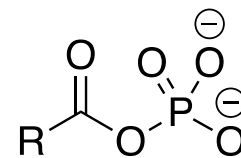
amide



ester

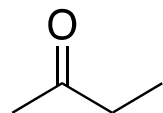
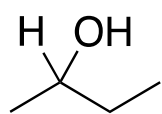


acyl chloride



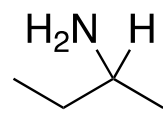
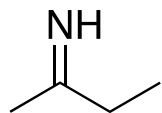
acyl phosphate

- **Oxidation** is the loss of electrons and **reduction** is the gain of electrons—still true in organic chemistry!
- We focus on the electron density at carbon atoms
- Oxidation involves the replacement of **C-H or C-C bonds** with bonds to **more electronegative heteroatoms** (especially in groups 15 – 17)



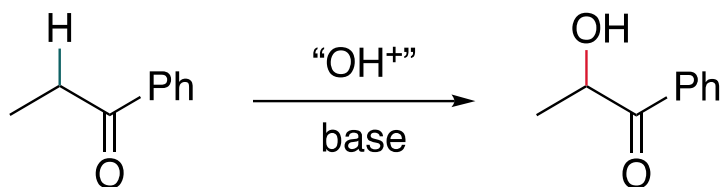
For oxidizing agent [O], this is typically $\text{H}^+([\text{O}]\text{H})^-$.

- Reduction involves the replacement of **bonds to electronegative heteroatoms C-X** with **C-H or C-C bonds**

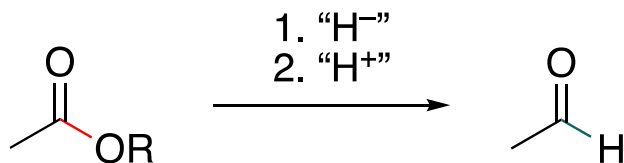


The actual reducing agent is typically “H^- ” followed by acid treatment (H^+).

- Oxidation and reduction may also involve the formation or cleavage of sigma bonds
- Oxidation involves the substitution of C-H or C-C bonds with C-X bonds
- Proton transfer followed by S_N2 or other substitution mechanisms are often involved



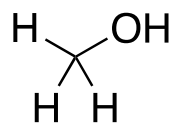
- Reduction involves the substitution of C-X bonds with C-H or C-C bonds
- Addition-elimination or S_N2 of a leaving group X with hydride (H⁻) are often involved



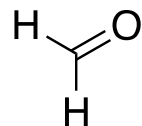
- Addition of hydrogen to an alkene or alkyne is reduction; the reverse is oxidation

12.4 Assigning Oxidation Numbers

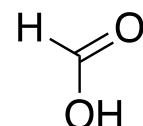
- To assign oxidation number, push *all* bonding electrons to more electronegative atoms and examine the resulting formal charges
- Carbon is more electronegative than hydrogen!
- C-X bonds (X = more electronegative heteroatom) increase the oxidation number of carbon



-2

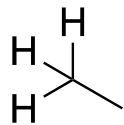


0

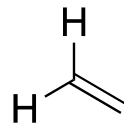


+2

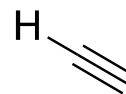
- Carbon-metal and carbon-hydrogen bonds decrease the oxidation number of carbon



-3



-2

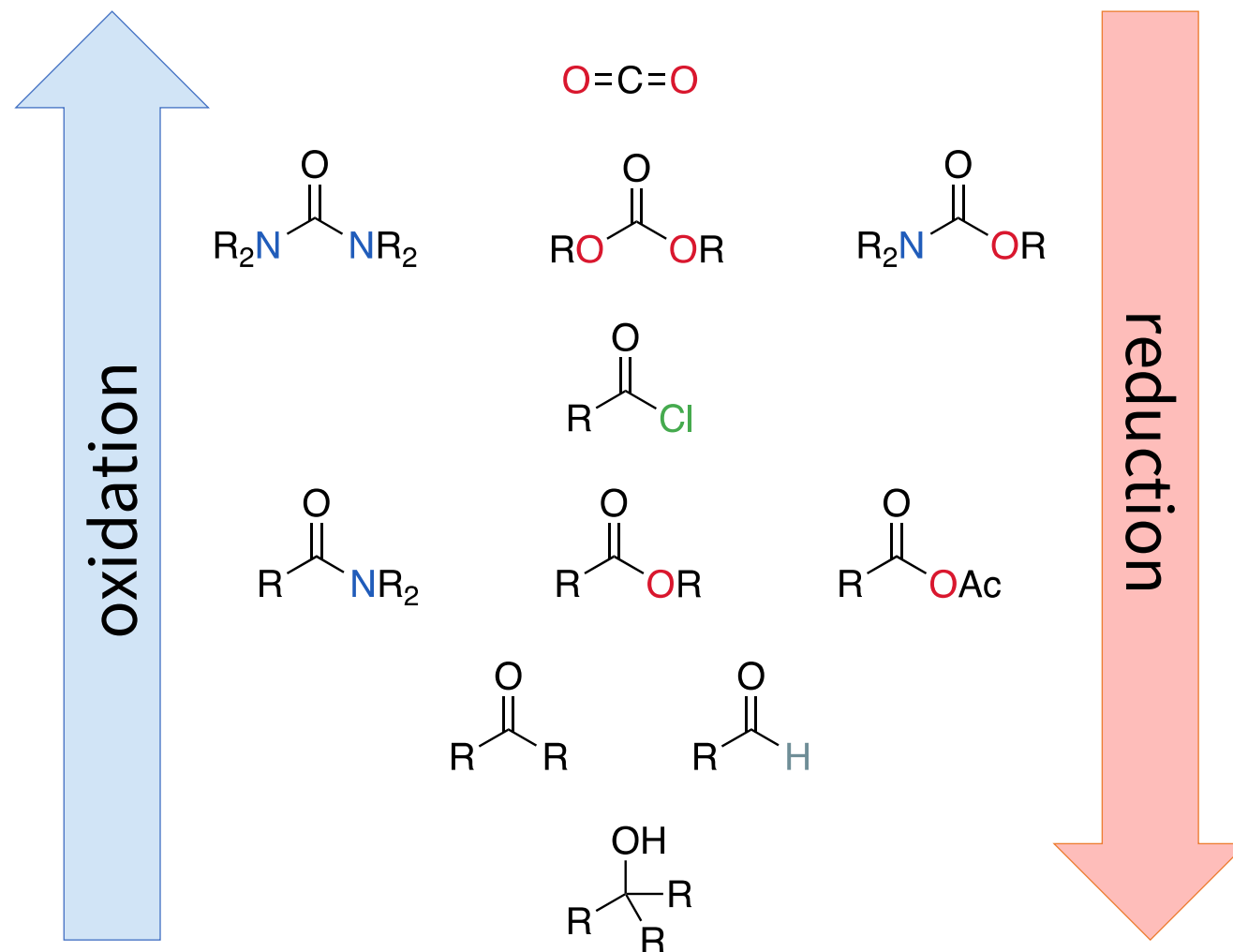


-1

- Note on the previous slides that oxidation involves an increase in oxidation number and reduction a decrease

12.4 Oxidation Ladders

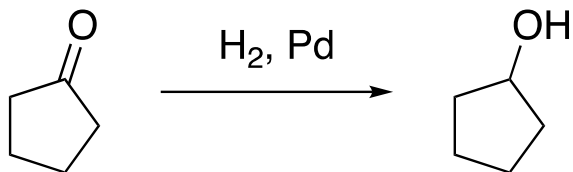
- It is useful to think of carbonyl compounds and alcohols on an **oxidation ladder** from most to least oxidized
- Use oxidation ladders to organize organic redox reactions
- Moving up, C-H or C-C is replaced with C-X
- Moving down, C-X is replaced with C-H or C-C
- “Horizontal” transformations are just substitution reactions with no change in oxidation number
- A similar ladder is useful for alkynes, alkenes, and alkanes



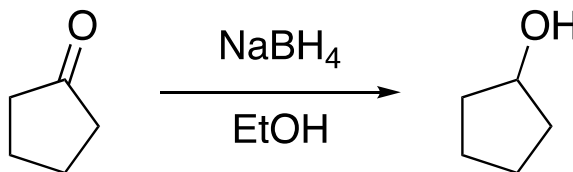
12.4 Reductions of Ketohydes to Alcohols

- Treatment of a ketone or aldehyde with a reducing agent yields an alcohol

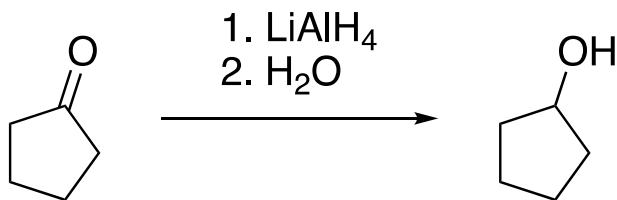
1. Hydrogen gas with a metal catalyst (rarely used)



2. Sodium borohydride (NaBH_4) – a **complex metal hydride**

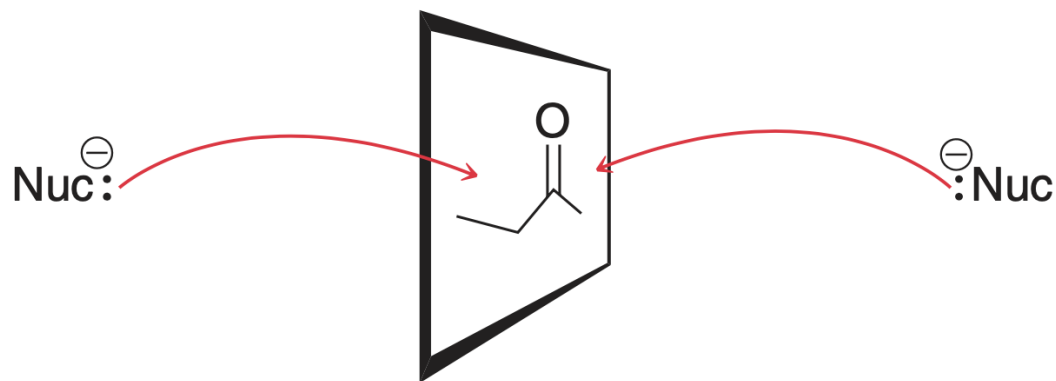


3. Lithium aluminum hydride (LiAlH_4) – a much more reactive reagent



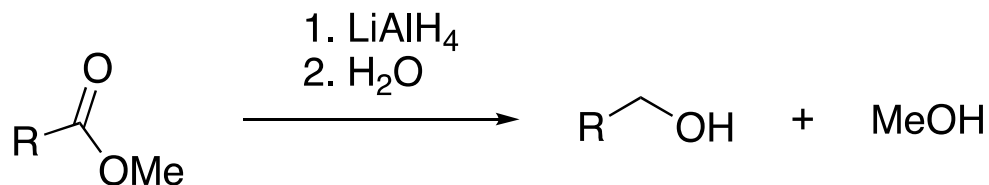
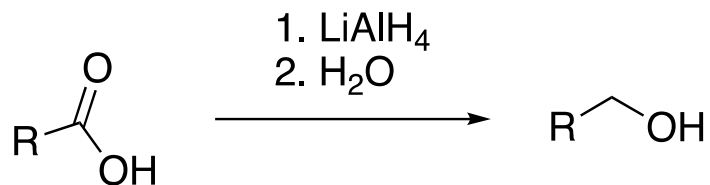
12.4 Mechanism of Complex Metal Hydride Reductions of Ketohydes

- Nucleophilic addition of hydride to the carbonyl carbon (Ad_N) followed by proton transfer to the carbonyl oxygen (pt)
- When a chiral product is formed from an achiral starting material, a racemic mixture of enantiomers is the result



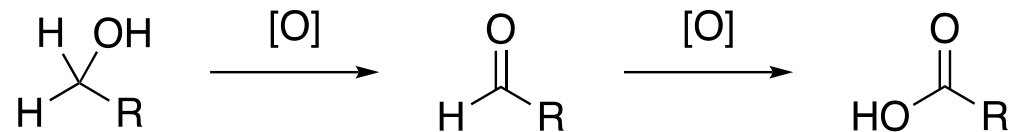
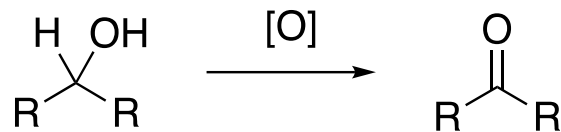
12.4 Reduction of Esters and Carboxylic Acids with LiAlH_4

- Esters and carboxylic acids are reduced to alcohols via two additions of hydride



- Mechanism involves an intermediate ketohyde formed via addition-elimination, a pattern we will see again soon

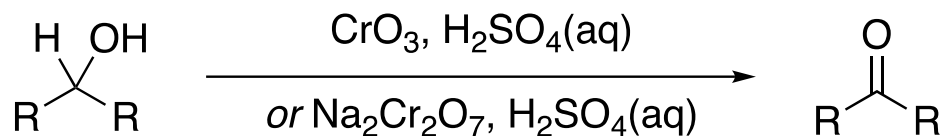
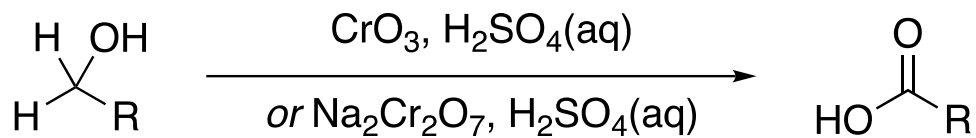
- Secondary alcohols are oxidized to ketones; primary alcohols to either aldehydes or carboxylic acids



- Tertiary alcohols are unreactive under oxidative conditions...why?
- The general concept of oxidation: installation of a nucleofuge on the carbonyl oxygen and elimination

12.10 Chromic Acid is a Strong Oxidizing Agent

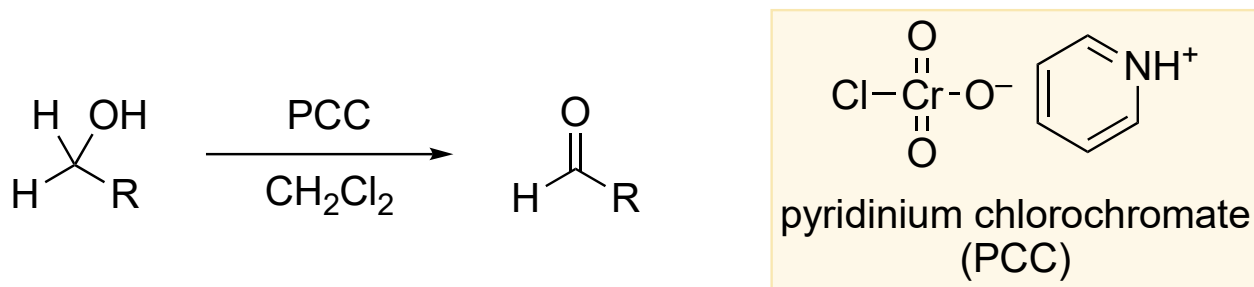
- Aqueous chromic acid (H_2CrO_4) is a **strong oxidizer** that converts primary alcohols all the way to carboxylic acids



- The mechanism involves formation of a chromate ester (1. pt; 2. $\text{S}_{\text{N}}2$) followed by elimination (3. E2)
- Intermediate aldehyde undergoes hydration followed by a second round of the mechanism above

12.10 Partial Oxidations: Pyridinium Chlorochromate (PCC) and Swern Oxidation

1. Anhydrous pyridinium chlorochromate (PCC) in CH_2Cl_2 converts primary alcohols selectively to aldehydes

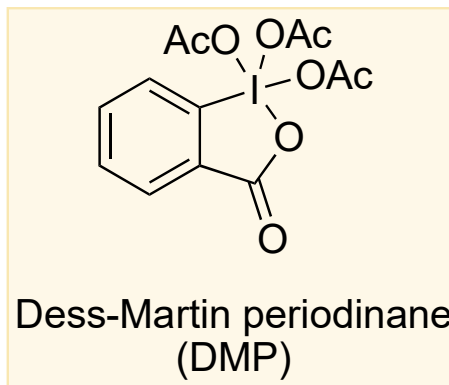
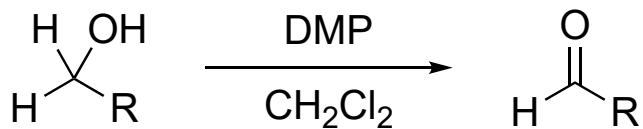


2. Swern conditions also convert primary alcohols selectively to aldehydes; the mechanism is instructive!



12.10 Partial Oxidations: Dess-Martin Periodinane

3. Dess-Martin periodinane (DMP) is one more reagent for conversion of primary alcohols to aldehydes

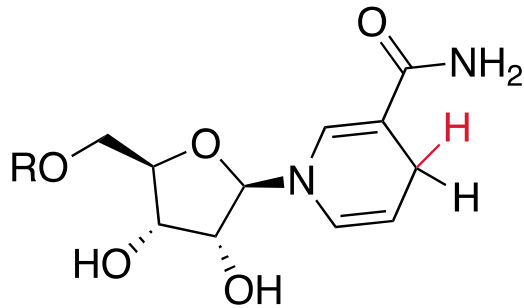


- All three of the partial oxidants are completely interchangeable for our purposes. Pick your favorite!

12.11 Nature's Reducing and Oxidizing Agents | 1 of 2

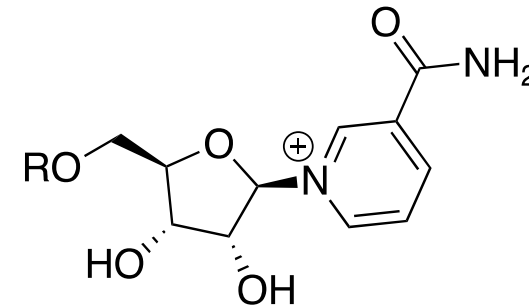
- Biochemical oxidation and reduction are mediated by hydride acceptors E^+ and hydride donors EH , respectively
- NADH, NADPH, $FADH_2$, and QH_2 are reducing agents
- NAD^+ , $NADP^+$, FAD, and Q are oxidizing agents

Reducing agent (reduced form)

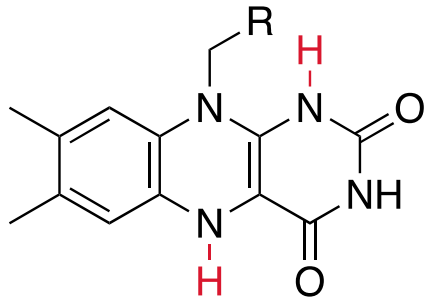


nicotinamide adenine dinucleotide (NADH)

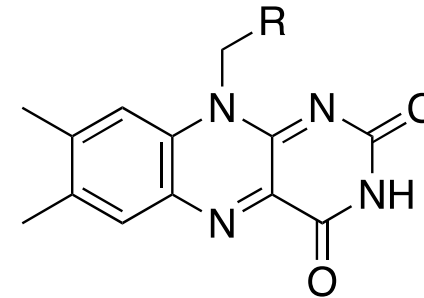
Oxidizing agent (oxidized form)



nicotinamide adenine dinucleotide (NAD⁺)



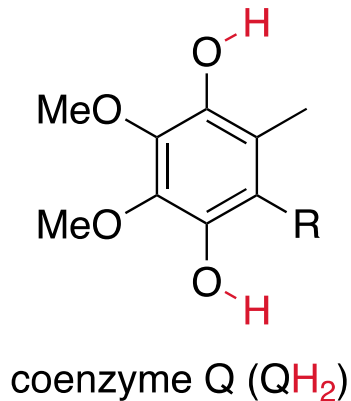
flavin adenine dinucleotide (FADH₂)



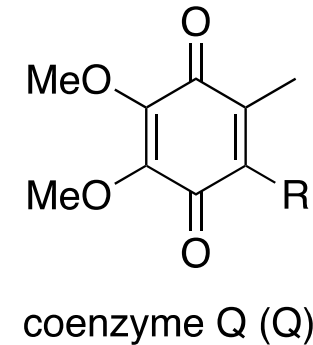
flavin adenine dinucleotide (FAD)

- Biochemical oxidation and reduction are mediated by hydride acceptors E^+ and hydride donors EH , respectively
- $NADH$, $NADPH$, $FADH_2$, and QH_2 are reducing agents
- NAD^+ , $NADP^+$, FAD , and Q are oxidizing agents

Reducing agent (reduced form)

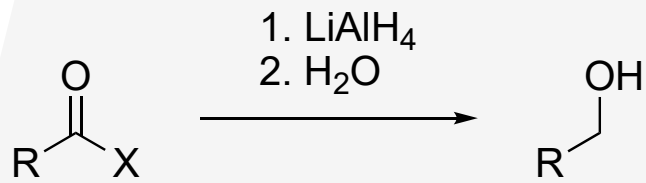


Oxidizing agent (oxidized form)

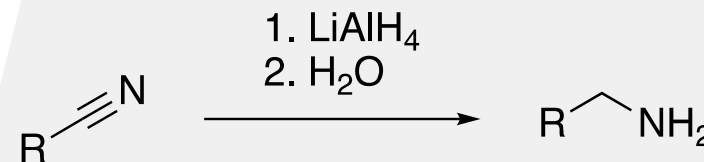
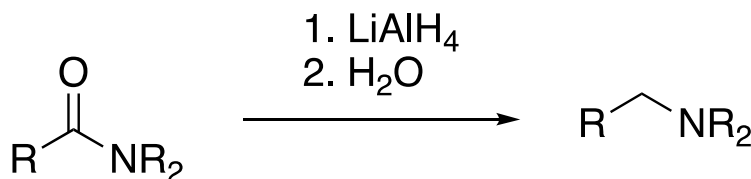


Reductions of Carboxylic Acid Derivatives

- Lithium aluminum hydride reduces esters, anhydrides, thioesters, and acid chlorides to alcohols
- Amides and nitriles are reduced to amines by lithium aluminum hydride



$\text{X} = \text{OR}, \text{OCOR}, \text{SR}, \text{Cl}$



- Diisobutylaluminum hydride (DIBAL-H, $i\text{-Bu}_2\text{AlH}$)** is a less reactive reagent that reduces esters, acid chlorides, and nitriles to aldehydes

